

Predicting Purchasing Outcomes from Online Shopping Sessions

Classification Question

How accurately can browsing behavior and session information distinguish online shopping sessions that result in a purchase from those that do not?

The project aims to investigate patterns associated with conversion in online retail using machine-learning classification.

Dataset

The **Online Shoppers Purchasing Intention Dataset** from the UCI Machine Learning Repository, contains 12330 sessions, 17 input features and the binary target **Revenue**. The target variable is Revenue, which records whether a session resulted in a purchase.

Approximately 15.5% of sessions resulted in purchase.

Candidate features include page-visit counts, time spent on pages, bounce and exit rates, visitor type, month, weekend status and several coded session attributes. Numeric codes representing categories will be treated as categorical rather than continuous variables.

Although UCI reports no missing values, I will check missingness, duplicate records, unusual values and the class distribution before modelling.

Sakar, C. & Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5F88Q>.

Propose Approach

A majority-class dummy classifier will establish a baseline performance. Logistic regression and decision tree will then be compared.

The dataset will be divided using a stratified train-test split because of the class imbalance.

Cross-validation will be performed on the training data. Purchase-class F1 will be the primary metric, supported by precision, recalls and accuracy.

Class weighting will be investigated as a possible method of addressing imbalance.

Responsible ML

The dataset does not contain any personal identifiers. However, the browsing behaviour could potentially be used by retailers for customer targeting. Results should therefore be treated as predictions rather than explanations of why individuals make purchase.

Repository: <https://github.com/Olszewski-Jakub/Online-Shoppers-Purchasing-Intention>